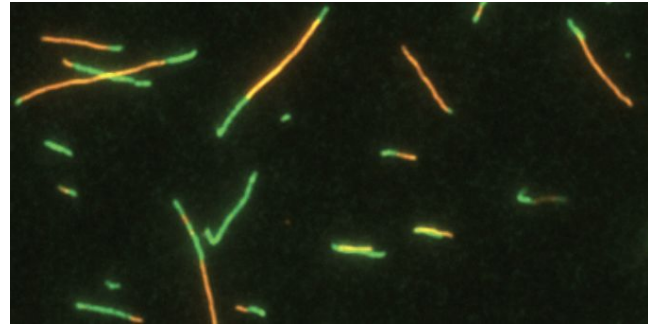
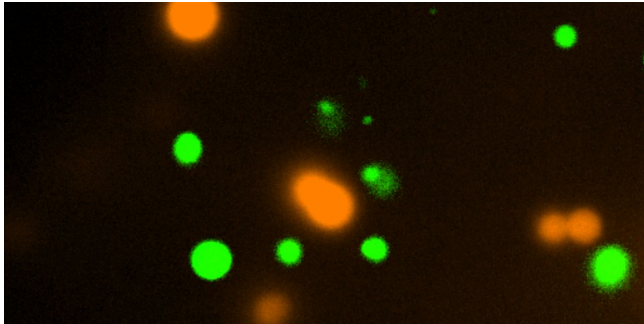


Automated Fluorescence Image Analysis of RNA Condensates

Daniel Chang

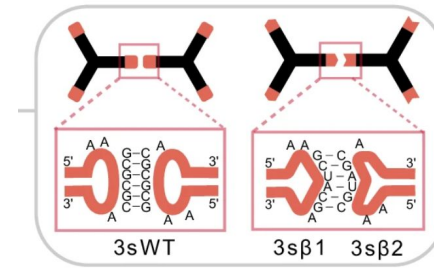
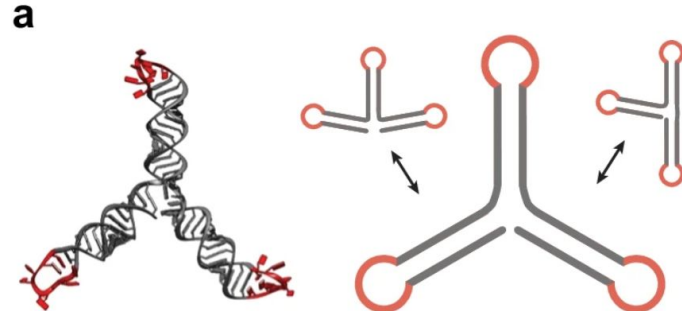
PI: Prof. Elisa Franco

Franco Lab | Dynamic Nucleic Acids Lab



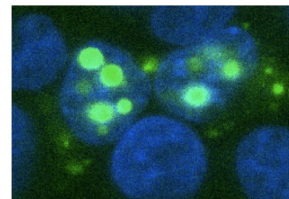
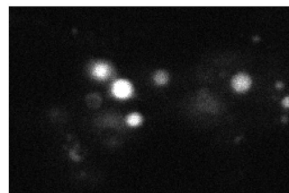
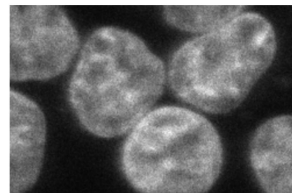
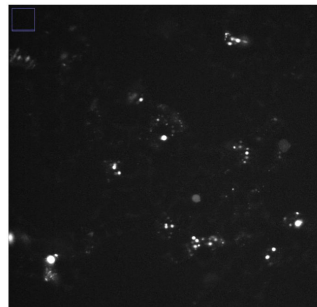
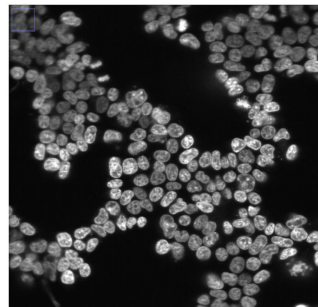
RNA Condensates

- **Condensates:** Dense, membrane-less droplets forming when molecules exceed a concentration
- Synthetic RNA condensates using programmable **RNA nanostars**
 - 100-200 nt single strands folding into multiarm structures and interact via kissing loops
- **Design Parameters:** arm length, number of arms, kissing loop sequence control



The JABr Construct: Dataset Explanation

- **JABr** = 15-nt stem, 3-arm, kissing-loop variant A (UCGCGA), Broccoli fluorescent aptamer
- 15-nt arm arm length = **nuclear pore permeability threshold** - forms in both nucleus and cytoplasm
- **Partition Coefficient (PC)** = Phase Separation Strength
 - Condensate intensity inside nucleus / intensity outside



The Problem: Manual Analysis Doesn't Scale

- **Existing analysis** was Manual, Slow, and Hard to reproduce
 - Prior: Manually check and segment nuclei and condensates using ImageJ/Fiji
- **Goal:** Automated Python pipeline taking in raw-stacks as inputs and outputs segmentation masks, measurements, and PCs

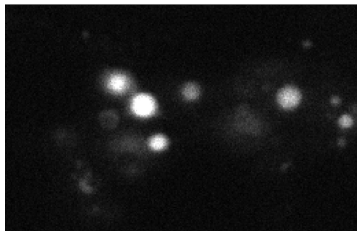
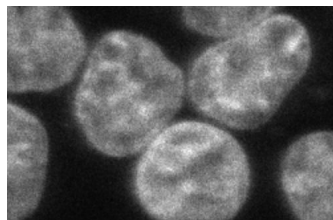
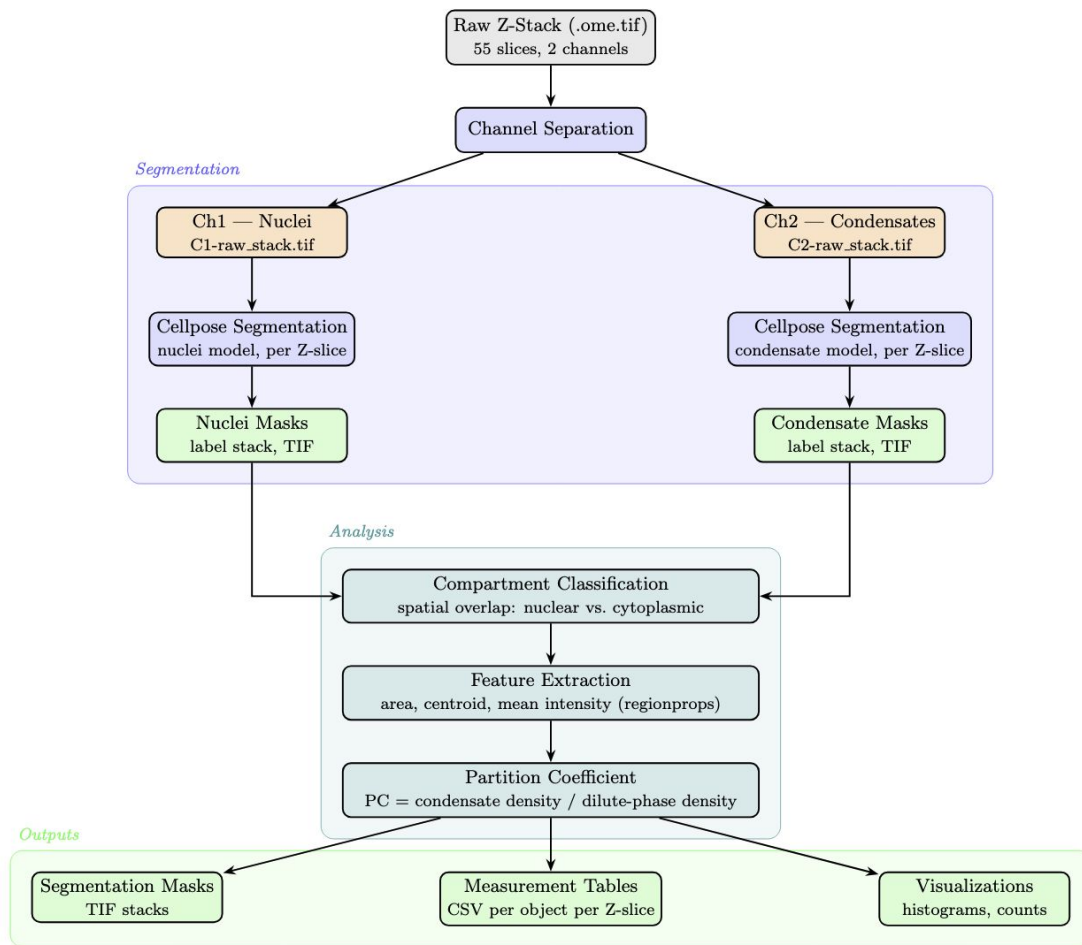
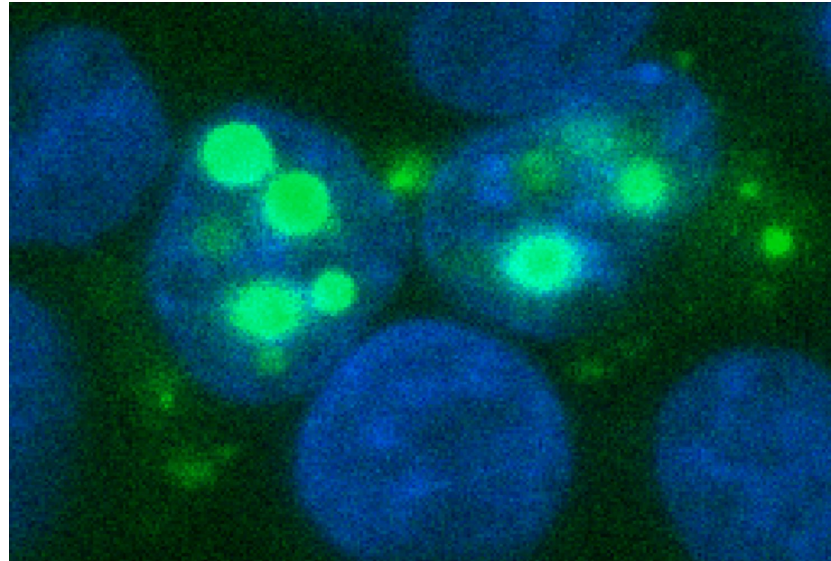


Image Analysis Pipeline



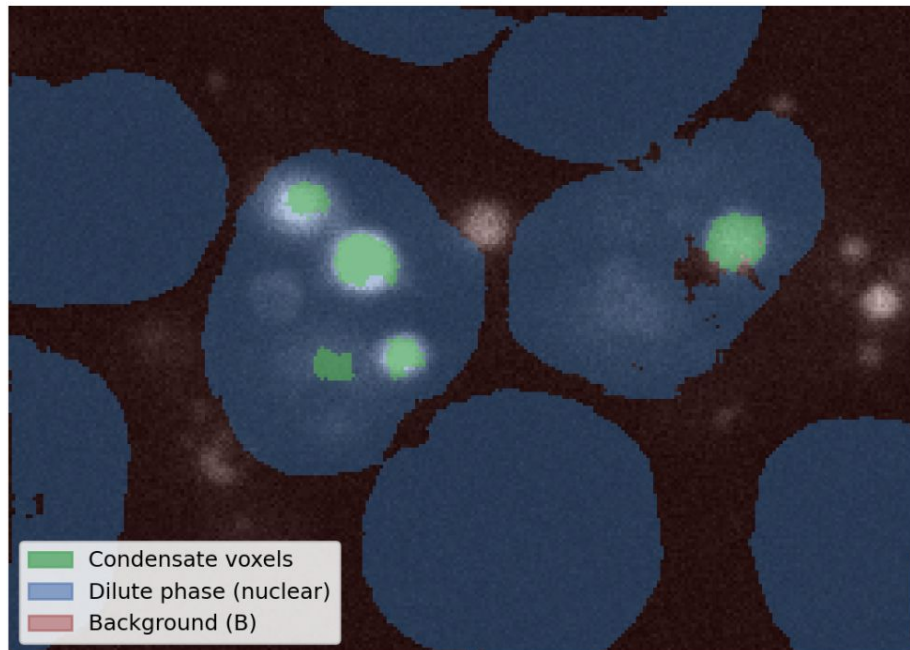
Quantifying Nuclear Enrichment

- **Partition Coefficient (PC)** = condensate density / dilute phase density
- **Condensate density** — mean intensity of pixels inside nucleus $\cap$ condensate mask
- **Dilute-phase density** — mean intensity of pixels inside nucleus, outside condensate mask
- **PC > 1** $\rightarrow$ condensates are enriched relative to surrounding nucleoplasm



Background Subtracted Partition Coefficient

Full slice (z = 20)



Partition Coefficient Formula

Step 1 — Background subtraction

$$B = \min(\text{intensity across full FOV})$$

removes camera/autofluorescence offset

Step 2 — Condensate density

$$\rho_{\text{cond}} = \text{mean}[\text{clip}(\text{pixel} - B, 0)]$$

over nuclear condensate voxels

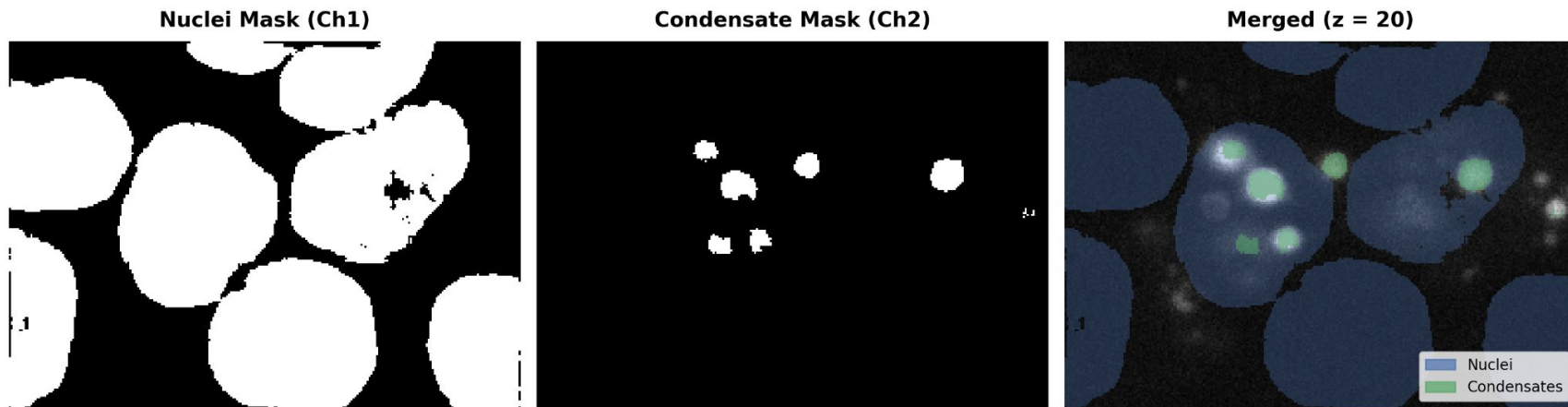
Step 3 — Dilute phase density

$$\rho_{\text{dil}} = \text{mean}[\text{clip}(\text{pixel} - B, 0)]$$

over 10×10×10 voxel patch in nuclear dilute phase

$$\text{PC} = \rho_{\text{cond}} / \rho_{\text{dil}} = 6.775$$

Segmentation Masks from Pipeline



- 80 nuclei and 186 condensates detected across 55 Z-slices
- Condensates localize predominantly inside nuclei — consistent with JABr nuclear enrichment
- Merged view confirms spatial overlap between condensate and nuclear masks

Measurement Tables from Pipeline

Condensate Measurements (cp3_condensate_measurements.csv)

label	area	y	x	mean_intensity	z
1	50	89.6	32.4	179.6	7
2	322	102.5	36.9	175.3	7
2	648	100.6	38.2	184.8	8
2	553	100.3	40.3	215.6	9
3	36	62.1	208.3	201.9	9
2	481	98.3	39.1	242.3	10
3	109	63.0	206.6	230.2	10
4	11	58.3	137.6	211.9	10

Nuclei Measurements (cp3_nuclei_measurements.csv)

label	area	y	x	mean_intensity	z
1	6	154.3	131.3	217.2	4
2	32	156.6	142.1	214.9	4
1	18	153.8	128.7	230.7	5
2	254	158.3	144.5	229.8	5
3	40	150.7	149.2	237.4	5
4	3	153.3	1.7	209.0	5
1	8	151.6	130.0	274.6	6
2	533	157.4	143.5	257.4	6

Segmentation Measurements from Pipeline

